

Types of Machine Learning

BEGINNER-FRIENDLY DEEP DIVE

Using simple examples to understand supervised, unsupervised, and semi-supervised learning.



The Big Idea

Machine Learning types depend on one key question:

Do we already have
the answer column in
the data?

✓ Answer Available

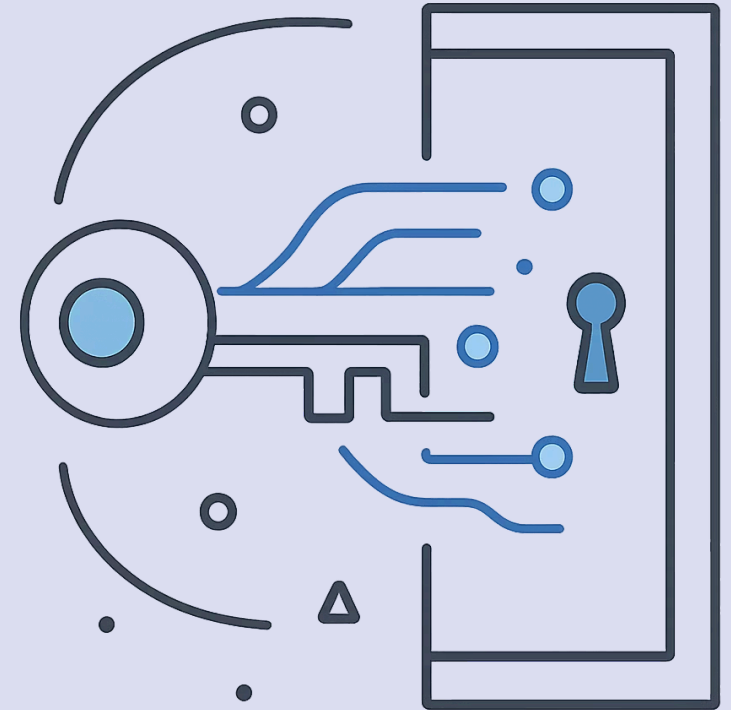
Supervised Learning

✗ No Answer

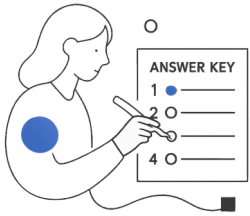
Unsupervised Learning

⚡ Some Answers

Semi-supervised Learning



The Three Main Types



Supervised Learning

Trains on data with correct answers already provided.



Unsupervised Learning

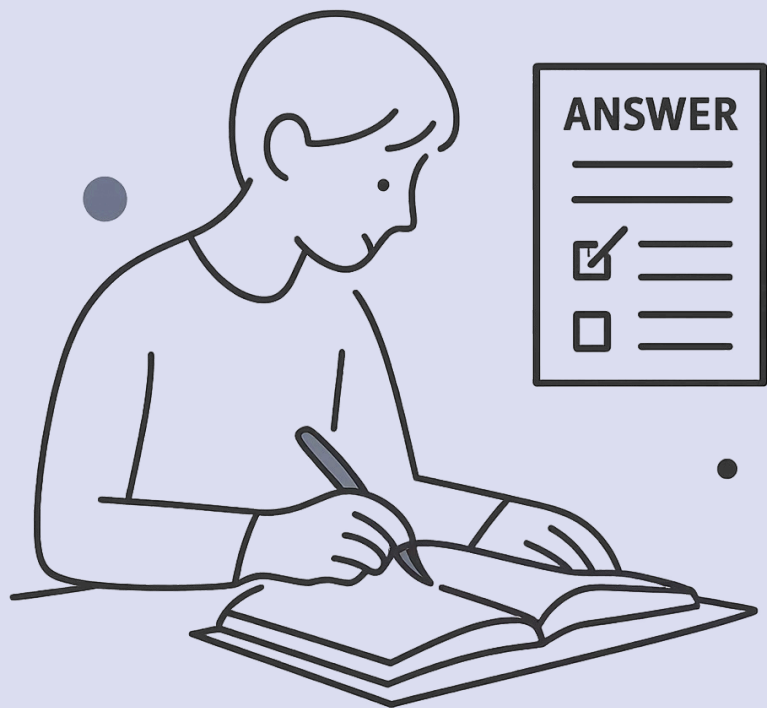
Finds hidden patterns in data with no answer column.



Semi-supervised Learning

Uses a mix — some labeled, mostly unlabeled data.

These types tell us how the machine learns from data — not just theory terms.



Supervised Learning

We train the machine using data where the correct answer is already available.

Think of it like a student learning with an answer key. The model studies question + correct answer pairs and learns the underlying pattern.

House Price Example

A classic supervised learning dataset — the Price column is the answer:

Area	Bedrooms	Age	Distance	Price
1000 sqft	2	5 yrs	8 km	₹55 lakh
1500 sqft	3	3 yrs	5 km	₹82 lakh
2000 sqft	4	2 yrs	4 km	₹120 lakh

 The model learns: Area + Bedrooms + Age + Distance → Price

Student Result Example

Study Hrs	Attendance	Prev. Marks	Result
2	60%	40	Fail
5	85%	70	Pass
6	90%	80	Pass
1	45%	25	Fail

Why Supervised?

The Result column is the answer — it's already provided in the training data.

The model learns:

📄 Study Hours + Attendance + Previous Marks → Pass or Fail

Two Branches of Supervised Learning

Regression

Predicts a number

House price, salary, temperature, delivery time

Classification

Predicts a category

Pass/Fail, Spam/Not Spam, Approved/Rejected



Supervised Learning

Learn from labeled examples

Regression

Predicts a continuous number

Classification

Predicts a discrete category



Regression

Predicting a number. When the output is continuous or numeric, it's Regression.



House Price

Input: Area, Bedrooms, Age → Output: ₹90 lakh



Salary Prediction

Input: Experience, Skills → Output: ₹12 LPA



Temperature

Input: Date, Location → Output: 32°C

Classification

Predicting a category or class. When the output is a label, it's Classification.



Pass or Fail

Input: Study hours, attendance → Output: Pass



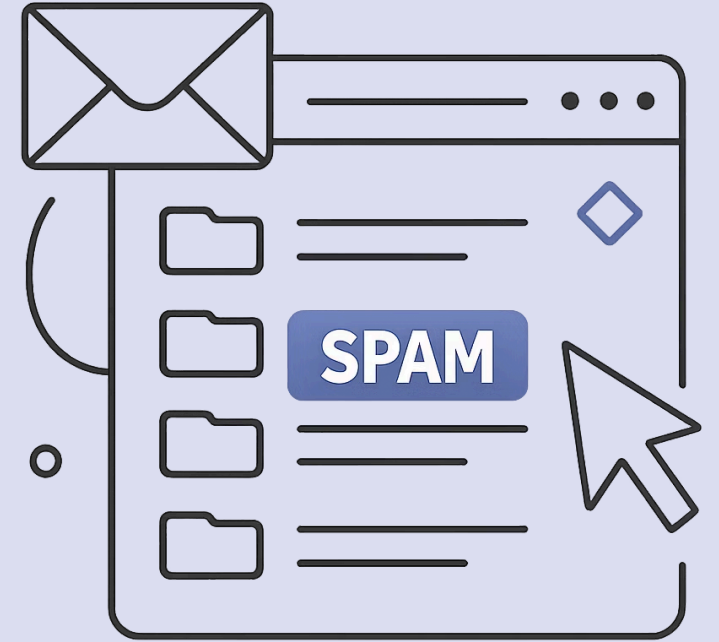
Spam Detection

Input: Email text, links → Output: Spam



Loan Approval

Input: Income, credit score → Output: Approved



Regression vs. Classification

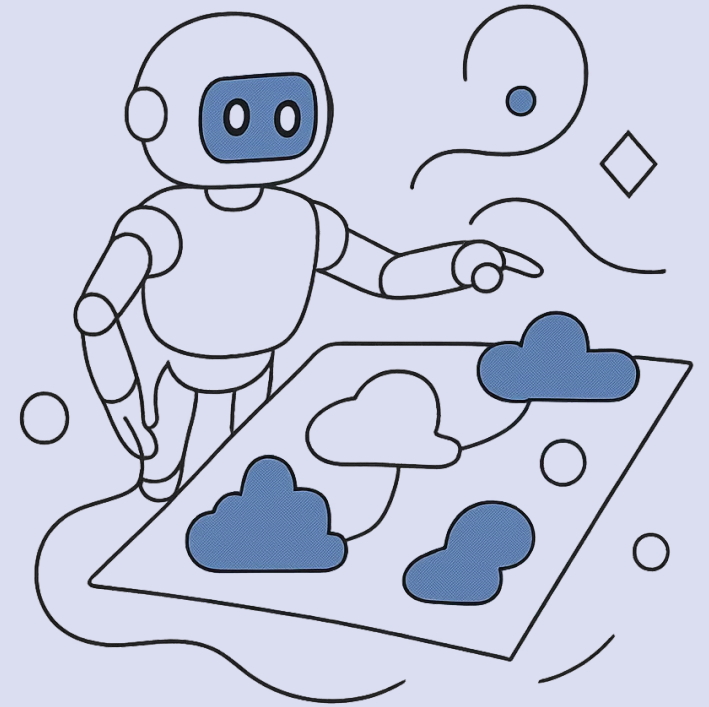
Question	ML Type
How much is the house price?	Regression
Will the student pass or fail?	Classification
What will tomorrow's temperature be?	Regression
Is this email spam?	Classification
Is this transaction fraud?	Classification

✔ Memory Rule: Number output → Regression | Category output → Classification

Unsupervised Learning

We give data to the machine, but there is no answer column. The machine finds hidden patterns by itself.

There is no teacher. No answer key. The model discovers structure on its own.



Customer Grouping Example

Customer	Age	Income	Spending
A	22	₹30K	High
B	45	₹1.2L	Medium
C	23	₹28K	High
D	50	₹1.5L	Low

No Answer Column!

There's no label like "Premium" or "Budget" customer. The machine groups similar customers automatically based on patterns it discovers.

Group 1

Young, high spending

Group 2

High income, low spending

Group 3

Middle income, medium spending

House Grouping: No Price Column

Area	Beds	Age	Distance
900	2	10	12 km
1100	2	8	10 km
2000	4	2	4 km
2200	4	3	5 km

Model Discovers Groups

- Small and far houses
- Large and near-city houses
- Medium houses

No price was given — the model found structure on its own.

Use Cases of Unsupervised Learning



Customer Segmentation

Group customers into discount seekers, premium buyers, frequent buyers, and window shoppers.



Market Basket Analysis

Find products frequently bought together to power recommendations.



Anomaly Detection

Identify unusual behavior patterns in transactions or usage data.



Supervised vs. Unsupervised

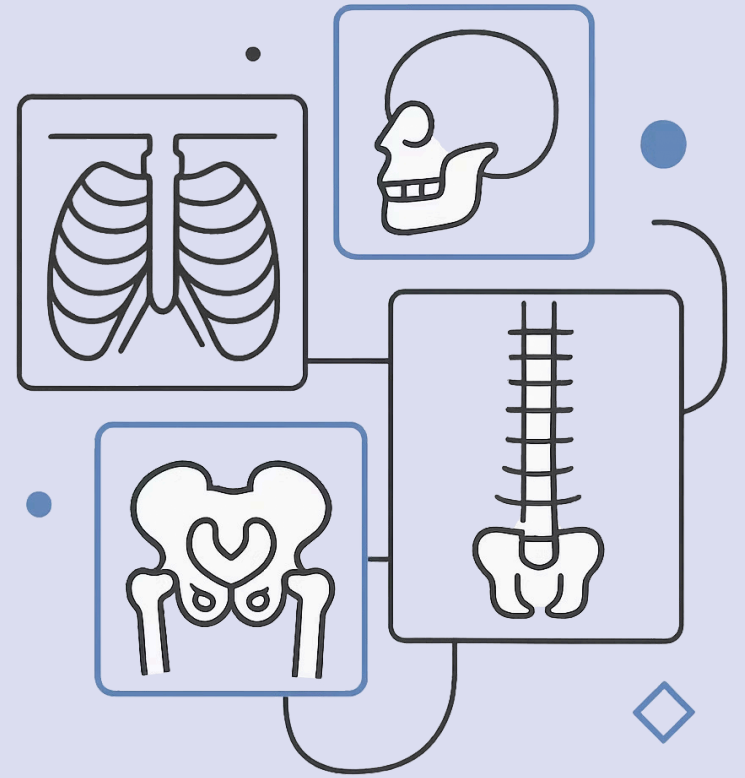
Point	Supervised	Unsupervised
Answer column?	✓ Yes	✗ No
Learns from correct answers?	✓ Yes	✗ No
Main goal	Prediction	Pattern discovery
Example	Predict house price	Group similar houses
Output	Known target	Hidden pattern

📌 Memory Trick: Supervised = With answer | Unsupervised = Without answer

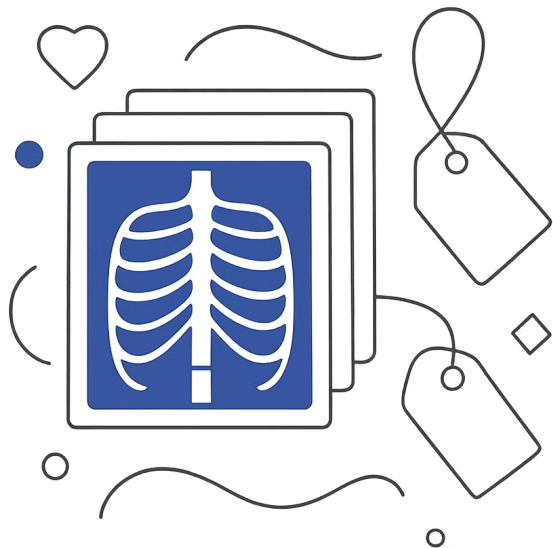
Semi-supervised Learning

Some data has answers, but most data does not. It sits between supervised and unsupervised learning.

The model learns from a small labeled dataset and a large unlabeled dataset together.



Medical Image Example



10,000 X-ray Images

Only 500 images are labeled by doctors. The rest are unknown.

Image	Label
Image 1	Normal
Image 2	Pneumonia
Image 4	Unknown
Image 5	Unknown

The model learns from both labeled and unlabeled data.

House Price: Semi-supervised

Area	Bedrooms	Age	Distance	Price
1000	2	5	8 km	₹55 lakh
1500	3	3	5 km	₹82 lakh
2000	4	2	4 km	Unknown
900	2	10	12 km	Unknown

 Some rows have price. Many rows do not. This is Semi-supervised Learning.

The Classroom Analogy

Supervised

Teacher gives questions and answers.

Q: $2 + 2 \rightarrow$ A: 4

Unsupervised

Teacher gives only objects.
Student groups them.

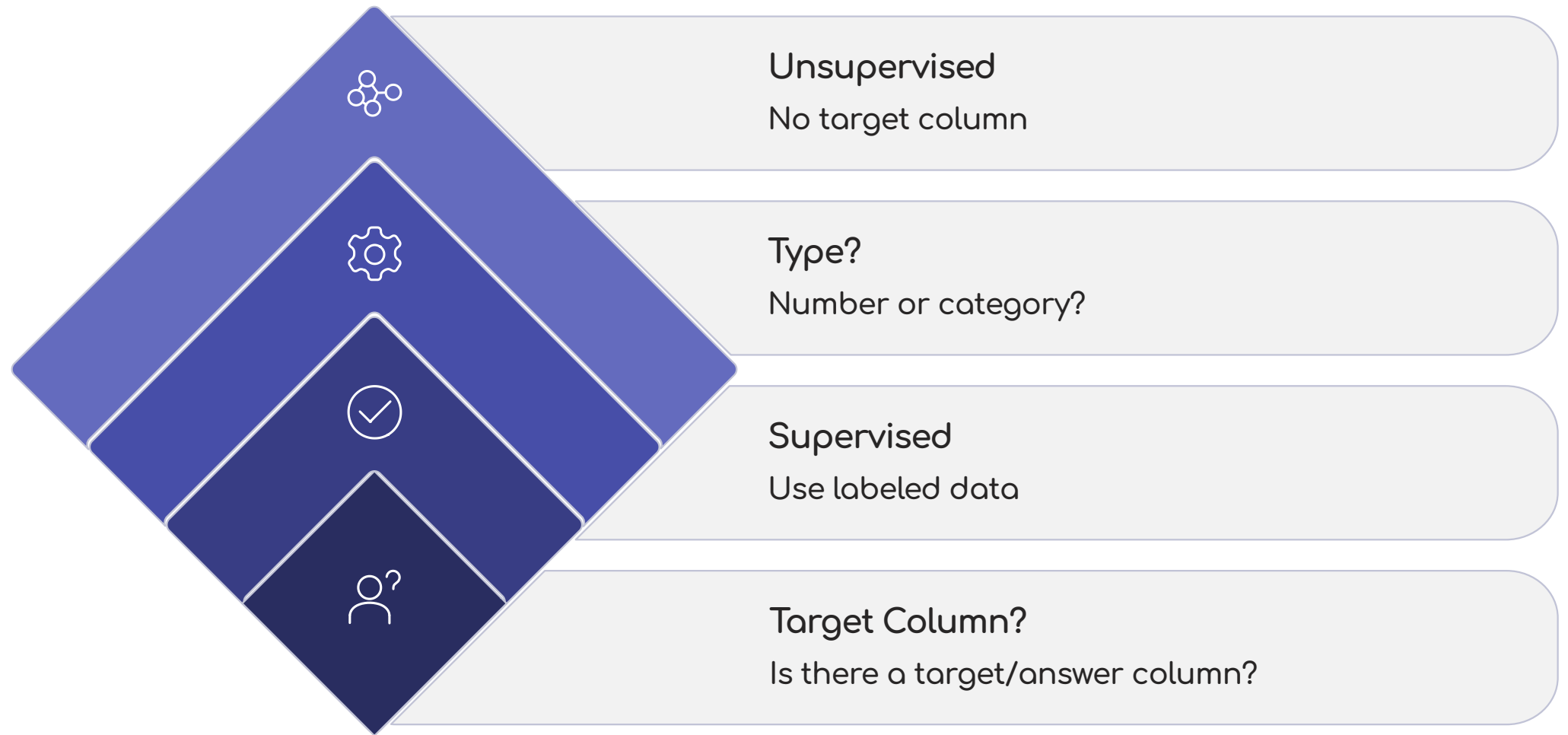
Apple, Banana, Carrot, Potato $\rightarrow$
student sorts

Semi-supervised

Teacher gives some answers only.

Apple = Fruit, Carrot = Vegetable,
Banana = ?

How to Identify the ML Type



This two-question framework lets you classify any ML problem quickly and confidently.

PRACTICE 1

Predict House Price

The Data

Area, Bedrooms, Location, Price

The Goal

Predict house price for a new listing.

Answer

Supervised Learning → Regression

Price is available in the data, and price is a number.

PRACTICE 2

Spam Detection

The Data

Email text, Sender, Links, Spam / Not Spam

The Goal

Predict whether a new email is spam.

Answer

Supervised Learning → Classification

The answer is available, and the output is a category.

PRACTICE 3

Group Similar Customers

The Data

Customer age, income, spending score

The Goal

Group similar customers together.

Answer

Unsupervised Learning

There is no answer column — the model discovers groups on its own.

PRACTICE 4

Learn Product Categories

The Data

10,000 product images. Only 500 images have labels.
Remaining images are unlabeled.

The Goal

Learn product categories from the dataset.

Answer

Semi-supervised Learning

Only some data has labels — the rest is unlabeled.

Common Confusions

Confusion 1

Supervised does not mean a human is watching the model. It means the training data contains correct answers.

Confusion 2

Unsupervised learning does learn — it learns hidden patterns without answer labels.

Confusion 3

Regression and Classification are inside Supervised Learning — they are not separate types.



Final Summary

Supervised

Data has answer column

Unsupervised

Data has no answer column

Semi-supervised

Some data has answers, most does not

Regression

Predict a number

Classification

Predict a category



Quick Quiz

A bank has old customer data: Age, income, credit score, loan amount, loan approved or rejected.

They want to predict whether a new customer's loan should be approved.

What type of ML is this?

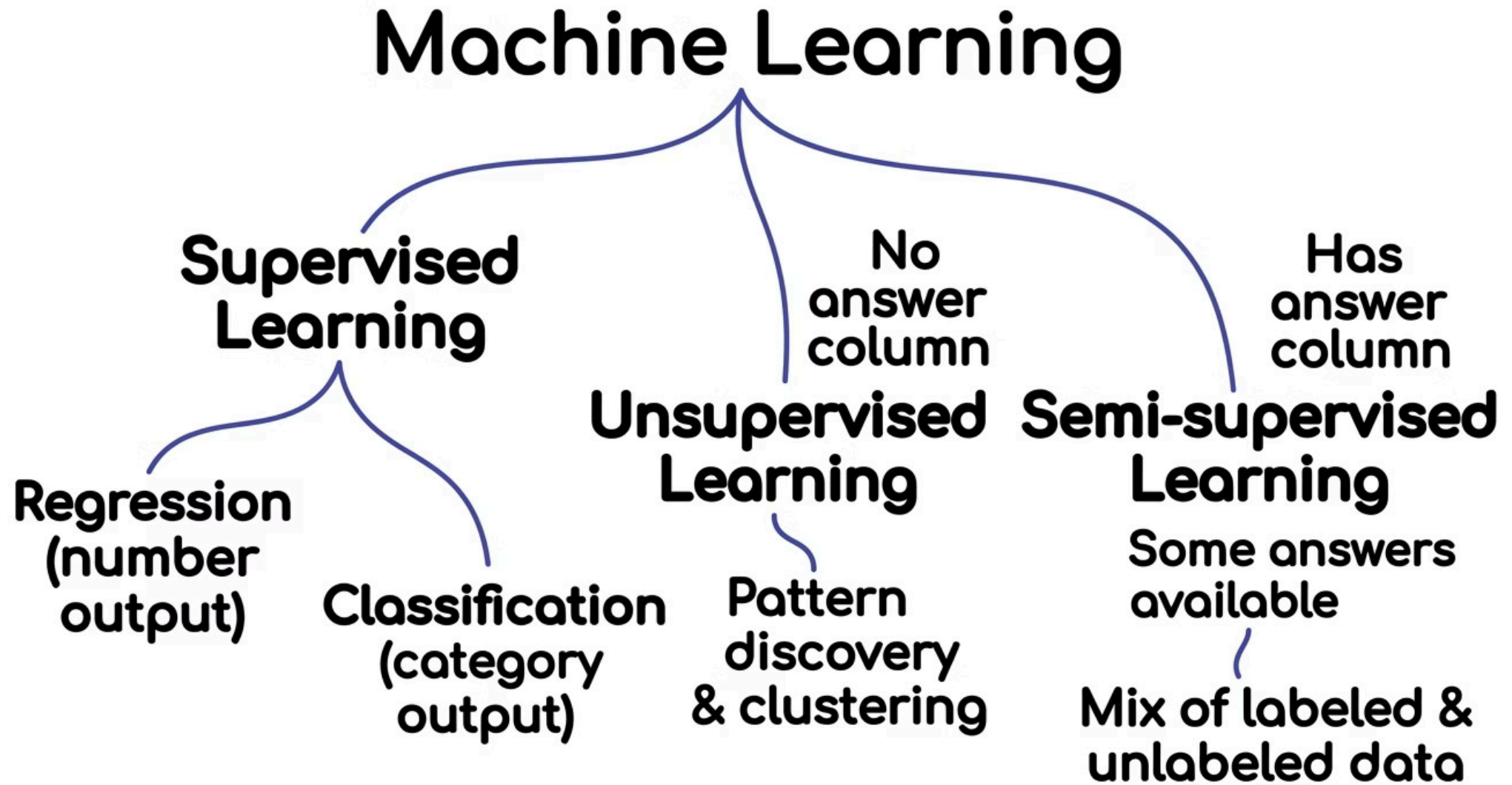
Quick Quiz Answer

Supervised → Classification

1 Answer column is available
The dataset includes "approved / rejected" — a known label.

2 Output is a category
Approved or Rejected — not a number. That makes it Classification.

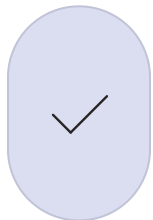
The Full Picture



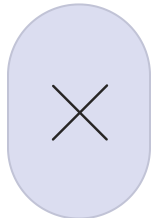
You've Got This!

You now understand the core framework of Machine Learning types.
Remember the one key question:

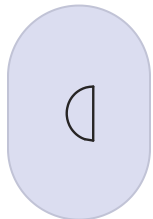
Is there an answer column?



Yes → Supervised



No → Unsupervised



Some → Semi-supervised

